

OLIVER ERNSTER

Technical & Engineering Leader

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Education

BSc (Hons) Physics, University of Manchester.

Final-year project:

Infra-red reconnaissance satellite design.

Decision Architecture (Published Work)

Author of a four-book Decision Architecture series and supporting site on how decision flow, authority boundaries and organisational design shape engineering outcomes.

The work translates directly into my engineering leadership: make ownership explicit, reduce decision latency and align architecture with the authority needed to operate it.

- Decision latency and coordination cost
- Authority boundaries and ownership clarity
- Structural failure modes in technical organisations

Available via Amazon and at:

<https://www.crankthecode.com/>

Senior Python engineer and technical lead with 28+ years in software engineering, including API/interface work throughout and 13+ years focused on Python backend, API and application-oriented systems.

I build and improve production services across defence, fintech, telecoms, cloud and startup environments, with a bias toward systems that are reliable, maintainable and straightforward to operate.

I also use agentic AI workflows - Roo/Cline with MCP, plus Claude, OpenAI and Google models - for code exploration, refactoring, tests, documentation and architectural option analysis while retaining ownership of design and production behaviour.

Core skills

- Python development: 13+ years building APIs, backend services, application tooling, automation and production systems; strong object-oriented and functional design-pattern experience in production Python.
- API/interface experience: 28+ years designing, consuming and integrating software interfaces across backend, application, telecoms and C/C++ systems.
- Cloud/platform: AWS Lambda, Zappa, AWS CDK, Terraform, Docker, CI/CD, deployment automation and production hardening.
- Modernisation: legacy migration, Python 2 to 3, dependency rationalisation, API replacement and maintainability.
- Leadership: technical direction, stakeholder alignment, delivery planning and multidisciplinary team leadership.
- Systems foundation: 4 years C and 15 years C++ from earlier career work; substantial object-oriented design, functional decomposition, interface design and multithreaded systems experience.
- Decision-aware architecture: ownership boundaries, lower coordination cost and clearer routes from design to operation.

Portfolio

Project portfolio: <https://www.crankthecode.com/portfolio>

- Public systems include NarrateX, Calendifier, LatencyLab, EDColonisation Assistant, Trainer and 3D-printing knowledge tools.
- Evidence spans cross-OS desktop apps, web systems, data/operational tools and hardware-adjacent work; Calendifier demonstrates Home Assistant integration.

Career history

Cisco - Senior Python Developer (Consultancy) - June 2025 to November 2025

- Accelerated platform modernisation, refactoring and architectural iteration through agentic AI-assisted workflows while preserving production accountability.
- Improved maintainability and release predictability through platform upgrades, dependency rationalisation and idempotent deployment practices.
- Improved usability, stakeholder confidence and platform operability by establishing a modern React/TypeScript interface.

Harrier - Lead Python Developer (Startup) - November 2024 to February 2025

- Delivered a serverless Python backend supporting ingestion, processing and reporting without requiring standing application infrastructure.
- Made AWS operation repeatable, secure and scalable by defining the infrastructure in Terraform from first principles.
- Reduced early-stage technical risk by owning architecture, system behaviour and reliability decisions under startup constraints.

LTI Mindtree - Lead Python Developer (Consultancy) - February 2023 to August 2023

- Reduced operational complexity and cost by replacing a MuleSoft API with a maintainable FastAPI Python 3 platform.
- Improved long-term maintainability and team handover through clearer architecture, coding standards and documentation practices.
- Preserved continuity for downstream consumers while moving the API onto a simpler delivery and support model.

EPAM - Lead Developer (Consultancy) - August 2022 to November 2022

- Gave stakeholders a concrete route for decision-making by prototyping a cloud-based flood insurance API in a regulated environment.
- De-risked the proposed architecture across FastAPI, Lambda/serverless, PostgreSQL, SQLAlchemy, AWS CDK and TypeScript.

MDRxTech - Senior Developer - January 2022 to July 2022

- Supported evolution of music-technology services by delivering Python/FastAPI backend work across Docker, serverless/Lambda and SQL-backed systems.

Thales - Senior Developer (Cyber Resilience & Defence Research) - September 2021 to December 2021

- Enabled cyber resilience modelling and visualisation for research users through a multithreaded socket-based backend and Flask front end.

Unilink - Senior Developer - March 2021 to September 2021

- Supported secure family communication and reoffending-reduction goals by delivering reliable Django/DRF software under operational, ethical and regulatory constraints.

Validus Risk Management - Senior Developer (FinTech) - October 2020 to March 2021

- Improved a regulated wealth management and risk analysis platform with enhancements aligned to analytical requirements and business objectives.

Memset - Senior Developer (Cloud Platform) - August 2019 to October 2020

- Strengthened core Python/Django cloud platform capabilities and contributed to platform-level architecture in a pure cloud engineering team.

Hewlett Packard - Senior Developer / Project Technical Lead / Scrum Master - December 2012 to February 2014

- Improved delivery coordination and predictability for a 17-person multidisciplinary defence team by owning technical direction, planning and Scrum practices.

Earlier career highlights

- Bink / Duologi / GlintPay: built Python/Django fintech startup experience across loyalty, lending and payments, adding regulated customer-facing financial systems depth.
- Armour Communications: strengthened secure-communications delivery experience through Python/Django/Linux/Docker work in a security-focused product environment.
- Anite: progressed through integration, support, development, project management and team leadership across 10 years in mobile telecoms.
- Northrop Grumman / DERA: applied C/C++ and Linux experience to defence and research systems, including multithreaded and protocol-oriented work.

Broader domain breadth

- Media and broadcast/editing: MPC Advertising, Forbidden and EditShare added experience across Python/Django, C++, Objective-C/Qt and production media workflows.
- Telecoms and public-sector systems: Semafone, Malden and NGO/Home Office work covered SIP/ISDN, C++/Python, search, Django and Linux delivery.
- IoT, art/startup and EV contexts: PhotonStar, Artfinder and Electric Motor Works broadened platform experience across IoT, Redis/Django systems and testing-heavy environments.
- Testing and delivery depth: module, system, integration, TDD, Selenium-driven web testing, conformance/interoperability testing, Jenkins, git hooks and standards-facing work.